KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 29 · MATH

The Tiny House Workshop

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Measuring Small Spaces

Liora is helping at a tiny-house workshop in Austin, Texas, where many measurements are fractions of a foot. Warm up before the lesson, and show your work. Some problems may feel new, and that is okay!

1. Review (Week 28): a) 0.6 × 0.9; b) 4.8 ÷ 0.6

2. Without multiplying, is 3/4 × 20 more or less than 20? Explain how you know.

3. Multiply: 2/3 × 3/5. Write your answer in simplest form.

4. Which number equals 1 1/2 × 8?

   12 | 9 1/2 | 4

5. Draw a square that stands for 1 square foot. Split it into 2 columns and 3 rows. What fraction of a square foot is each small piece?

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 29 · MATH

The Tiny House Workshop

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Ms. Linnea's Final Plans

Compare each factor with 1 to predict, and use four partial products for mixed numbers.

1. Without multiplying, write >, <, or =. a) 5/6 × 18 _____ 18; b) 9/7 × 18 _____ 18; c) 3/3 × 18 _____ 18

2. Find each area. a) a shelf 3/4 foot by 2/5 foot; b) a door panel 2 1/2 feet by 1 1/5 feet

3. Two steps: a main floor is 8 feet by 22 1/2 feet. The loft is 2/5 as large as the main floor. How many square feet do the main floor and loft have together?

4. Find the mistake: Lochlan says 2 1/2 × 2 1/2 = 4 1/4, because 2 × 2 = 4 and 1/2 × 1/2 = 1/4. In two or three complete sentences, explain his mistake and give the correct product.

★ **Challenge:** Draw a unit square and tile it to show 2/3 × 1/2. Then explain why the product is less than both factors.

Looking Ahead: Week 30: the Final Checkpoint. It brings together skills from the whole program, so look back at your Unit 6 notes on expressions, powers of 10, decimals, and fractions.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 29 · READING

Rolling Toward a Decision

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Taking a Side

Today you will judge two speakers at a Flagstaff City Council meeting who disagree about where to build a new skate park. Show what you already know. It is fine to guess!

1. Review (Week 28): Source A says a lake is "about 40 feet deep." Source B says it is "nearly 50 feet deep." Write one sentence that reports what both sources say and cites them.

2. Think of a time you tried to convince someone to agree with you. What did you want, and what reason did you give?

3. Which sentence is a **claim** someone could argue about?

   Flagstaff is in Arizona. | Our town should build a skate park. | The park closes at sunset.

4. Read: "Everybody in my class wants a later bedtime, so you should say yes." Is this a strong reason? Explain why or why not.

5. Why might it help a writer to explain the OTHER side's point of view before answering it?

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 29 · READING

Rolling Toward a Decision

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Weigh Both Sides

Write in complete sentences with correct capitals, commas, and transitions.

1. Two steps. Name one strong piece of evidence from Nizhoni AND one from Mr. Nolan. Then explain which site's argument you think is better supported, and why.

2. Find the mistake. Nikhil wrote: "Skate parks always cause trouble, so no town should ever build one." Name the reasoning trap and rewrite the sentence fairly.

3. The survey found 248 of 400 people support a skate park. Explain why this number cannot tell the council WHICH site to choose.

4. Be the editor. Rewrite correctly. a) dear council members b) However the park could of opened sooner.

★ **Challenge:** Write 3 sentences to the Flagstaff City Council for Site 1 OR Site 2: a claim, one reason, and a rebuttal to the other side.

Looking Ahead: Week 30: the Final Checkpoint. You will review the skills from the whole program, from myths and mysteries to research and persuasive writing.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 29 · SCIENCE

People Protecting Earth

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Week 28):** Name the three ways heat moves. Give one example of each.

2. Name one way people can harm the land, one way they can harm the water and one way they can harm the air.

3. What does it mean when an animal is *endangered*? Name an endangered animal you have heard of.

4. Predict: Since 1970, the United States economy and the amount of driving have grown a lot. Do you think air pollution went up or down? Why?

5. Draw one thing your family or community does to protect Earth. Label it.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 29 · SCIENCE

People Protecting Earth

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Evidence for a Healthier Earth

Use evidence from today's lesson. Write in complete sentences.

1. **Sort it.** Write land, water, air or living things for each: (a) a forest cleared for a mall, (b) CFCs thinning the ozone layer, (c) lead poisoning condors.

2. **Two-step.** A factory released 4,000 tons of a pollutant. It cut that amount by 85 percent. What percent is left? How many tons is that? Show both steps.

3. **Find the mistake.** Porter says, "The air got cleaner because nobody drives anymore." Use the EPA data to explain what is wrong.

4. **Explain.** Why does giving hunters non-lead ammunition work on the CAUSE of the condors' problem, not just the effect?

★ **Challenge: Environmental justice.** In your own words, explain what environmental justice means and why scientists like Robert Bullard collect data on where pollution is.

Looking Ahead: Next week: the Final Checkpoint. You will show what you have learned across the Pathways program. Bring your best thinking and your scientist habits!

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 29 · SOCIAL STUDIES

A Changing Nation (1965–Today)

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Write what you know right now. Guesses are fine; this is your starting point.

1. **Review (Week 28):** Name one event from the Cold War or the Space Race and tell why it mattered.

2. Why do you think families move to the United States from other countries? Write two reasons.

3. Prediction: Are the internet and the World Wide Web the same thing? Explain your guess.

4. What do you think Earth Day is for? Describe one way people could take care of a river or park.

5. You are not old enough to vote yet. Write one way a kid your age could still help a community.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 29 · SOCIAL STUDIES

A Changing Nation (1965–Today)

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: A Nation That Keeps Changing

Use evidence from today's lesson. Write in complete sentences.

1. **Put in order with years:** World Wide Web invented; first Earth Day; Immigration Act signed; 26th Amendment ratified; first ARPANET message.

2. **Find the mistake.** "The Immigration Act of 1965 kept the country quotas of 1924." Rewrite the sentence so it is true.

3. **Read the census.** In 1980, 14.1 million people in the United States were born in another country; in 2010, 40.0 million. By how many million did it grow? Which law helped cause this?

4. **Write a letter.** Write 3 or 4 sentences to your town council asking for one way to care for a local park or river. Give a reason and an example. Check capitals, commas and end punctuation.

★ **Challenge: Two steps:** Grace Hopper made the first compiler in 1952. The Web was invented 37 years later, and CERN made it free 4 years after that. In what year was the Web made free?

Looking Ahead: Next week: the Final Checkpoint. Look back at your Questions I Still Have lists and your Pre/Post sheets from Unit 6, and get ready to show what you know.